

CivicPulse AI

Predictive Civic Infrastructure Intelligence

Project Overview

CivicPulse AI is an explainable, AI-powered civic infrastructure intelligence platform designed to help Indian urban local bodies move from reactive complaint management to predictive and preventive maintenance.

Instead of waiting for citizens to report potholes, waterlogging, blocked drains, road damage, water leaks, or other civic failures, CivicPulse AI analyses available civic, weather, geographic, infrastructure, and population data to predict where a problem is likely to occur or become critical.

The platform then estimates the likely impact, identifies the probable cause, prioritizes the required action, and recommends the most effective use of limited municipal staff, time, equipment, and budget.

Final Project Direction

CivicPulse Monsoon

CivicPulse Monsoon is the focused version of CivicPulse AI.

It predicts ward-level drainage and waterlogging risks 24–72 hours before a likely failure, estimates the affected population and critical facilities, traces likely upstream causes through a road–drain–waterbody relationship graph, and generates a budget- and crew-constrained preventive inspection plan for municipal officers.

Core Problem

Indian urban local bodies commonly respond to waterlogging, blocked drains, road damage, and related civic-service failures only after citizens submit complaints or after heavy rainfall has already caused disruption.

Municipal information is fragmented across complaint systems, ward records, rainfall sources, maps, infrastructure departments, and field teams. Municipal authorities also operate with limited budgets, limited workers, incomplete infrastructure records, and insufficient real-time monitoring.

As a result:

- Problems are detected late.
- Emergency repairs are more expensive than preventive maintenance.

- Repeated failures may be treated as separate complaints.
- High-risk areas without active citizens may be overlooked.
- Departments may not identify the upstream infrastructure causing the visible problem.
- Available crews may be assigned without considering population impact, failure probability, travel time, or intervention effectiveness.

Proposed Solution

CivicPulse AI creates a predictive civic operations layer that combines:

- Historical civic complaints and incidents.
- Rainfall observations and forecasts.
- Road, drain, waterbody, and settlement information.
- Elevation and lowland or flow-accumulation indicators.
- Population density and critical-facility locations.
- Infrastructure age, recurrence, inspection, and repair information where available.
- Citizen reports, officer inspections, photographs, and optional sensor data.

The system produces:

1. A predicted infrastructure-risk map.
2. A 24-, 48-, and 72-hour risk forecast.
3. A probability and confidence level for each risk zone.
4. A list of evidence explaining the prediction.
5. An estimated population and critical-facility impact.
6. A likely upstream cause or affected infrastructure connection.
7. A prioritized preventive-action list.
8. A crew- and budget-constrained work plan.
9. A what-if simulation for rainfall and intervention scenarios.
10. A geospatial municipal operations dashboard.

Example Use Case

A ward contains an old drainage corridor near a densely populated settlement.

The system detects the following signals:

- Repeated waterlogging complaints in the same corridor.
- Increasing incident frequency during recent rainfall events.
- Heavy rainfall forecast during the next 48 hours.
- Low-lying terrain and poor drainage connectivity.

- A nearby blocked or undersized drain segment.
- High population exposure.
- A school and a primary healthcare centre near the predicted risk area.
- Previous repairs that did not prevent recurrence.

CivicPulse AI generates the following decision card:

- Infrastructure risk: High.
- Forecast window: Next 24–72 hours.
- Estimated exposed population: 8,000+; scenario-based estimate.
- Recurrence: High.
- Critical facilities nearby: One school and one healthcare facility.
- Main contributing factors: Rainfall anomaly, recurrence, low elevation, drainage proximity, and high exposure.
- Probable cause: Upstream drain blockage or insufficient drainage capacity.
- Priority: Critical.
- Recommended action: Inspect and clear the upstream drain before the predicted rainfall event.
- Confidence: Displayed separately from risk probability.
- Safety note: Field verification required before execution.

The platform must not display unsupported certainty. A value such as 87% should only be displayed when the model is trained, tested, and calibrated using reliable historical labels. Otherwise, CivicPulse should show a risk class, confidence level, evidence, and data-quality warning.

Main Objectives

- Predict civic infrastructure problems before complaints increase.
- Identify locations likely to experience waterlogging or drainage failure.
- Detect repeated and related incidents.
- Estimate the likely population and infrastructure impact.
- Identify probable upstream causes.
- Prioritize actions based on risk, impact, recurrence, and preventability.
- Optimize limited municipal crews and budgets.
- Support pre-monsoon preparedness and preventive inspection.
- Provide transparent and explainable AI recommendations.
- Reduce dependence on complaint volume alone.
- Support human verification and municipal accountability.

Target Users

- Municipal commissioners.
- Urban local bodies.
- Ward engineers.
- Stormwater-drainage departments.
- Public works departments.
- Water-supply departments.
- Disaster-management cells.
- Smart-city command centres.
- Field inspection teams.
- Municipal planning and budget officers.

Core System Workflow

1. Data collection

The system receives data from public government sources, municipal records, geospatial files, citizen reports, field inspections, and optional sensor feeds.

2. Data cleaning and standardisation

The system validates coordinates, timestamps, issue types, duplicate records, missing values, stale layers, and inconsistent department labels.

3. Civic-event understanding

Text, voice, images, and location data are converted into structured events containing:

- Issue category.
- Location.
- Time.
- Severity indicators.
- Affected asset or area.
- Evidence quality.
- Department ownership.

4. Incident and recurrence detection

Nearby reports occurring within a relevant time period are grouped into a single incident. Repeated incidents at the same location are identified as recurrence patterns.

5. Risk prediction

The model estimates the probability that a spatial cell, road segment, drainage area, or infrastructure zone will experience a critical civic issue during a future time window.

6. Impact estimation

The system overlays predicted risk with population, schools, hospitals, roads, settlements, and other sensitive locations to estimate potential exposure.

7. Root-cause and dependency analysis

The system analyses connected drains, roads, waterbodies, low-lying areas, outlets, and nearby infrastructure to identify likely upstream causes.

8. Priority calculation

Each predicted issue receives a priority score based on:

- Failure probability.
- Severity.
- Population exposure.
- Critical-facility exposure.
- Recurrence.
- Forecast rainfall.
- Infrastructure vulnerability.
- Preventability.
- Response time.
- Data confidence.

9. Resource optimisation

The system recommends which inspections or interventions should be performed first, considering crews, skills, travel distance, work hours, budget, ownership, and urgency.

10. Scenario simulation

Municipal officers can change rainfall intensity, crew availability, budget, blocked drains, road closures, or preventive actions to compare possible outcomes.

Recommended Risk Logic

A practical risk score can combine probability, severity, exposure, recurrence, and preventability:

$\text{Risk Priority} = \text{Failure Probability} \times \text{Impact} \times \text{Recurrence} \times \text{Preventability} \times \text{Data Confidence Adjustment}$

The system should show the components separately instead of hiding everything in one unexplained score.

Impact Estimation

Estimated impact can combine:

- Population within the predicted risk area.
- Schools and healthcare facilities.
- Important roads and transport links.
- Informal settlements and vulnerable communities.
- Water, sanitation, and public-service assets.
- Estimated duration of disruption.

The output must be described as estimated exposure, not a guaranteed number of affected people.

Key Differentiating Features

1. Pre-complaint risk forecasting

Predict risk before complaint volume increases by combining weather, recurrence, terrain, network, and exposure data.

2. Explainable risk cards

For every high-priority prediction, show why the model generated the result.

Example evidence:

- Rainfall anomaly is high.
- The location has repeated incidents.
- The area is low-lying.
- A drainage outlet is nearby.

- Population exposure is high.
- A critical facility is within the impact zone.

3. Drainage dependency graph

Represent relationships between drains, roads, waterbodies, settlements, outlets, and critical facilities.

This helps the authority inspect the likely upstream cause instead of repeatedly repairing only the visible downstream symptom.

4. Spatial risk propagation

Estimate how waterlogging or service disruption may spread along connected roads, drainage paths, and low-lying areas.

5. Recurrence intelligence

Identify locations where the same issue repeatedly returns after repair.

This helps distinguish a temporary symptom from a structural or upstream problem.

6. Complaint underreporting correction

A low number of complaints must not automatically mean low risk. CivicPulse adjusts interpretation using population, accessibility, reporting coverage, and independent geographic evidence.

7. Confidence-aware decision support

The system displays probability, confidence, data quality, and the recommendation to inspect when uncertainty is high.

8. Exposure-aware prioritization

An issue affecting a school, hospital, dense settlement, or important road may receive higher priority than an isolated issue with the same visual severity.

9. Crew allocation optimisation

The platform recommends a feasible schedule for available crews, considering travel distance, skills, budget, urgency, and expected risk reduction.

10. Intervention simulation

Officers can simulate the expected effect of clearing a drain, inspecting a corridor, closing a road, increasing crew capacity, or receiving heavier rainfall.

11. Multi-hazard relationship analysis

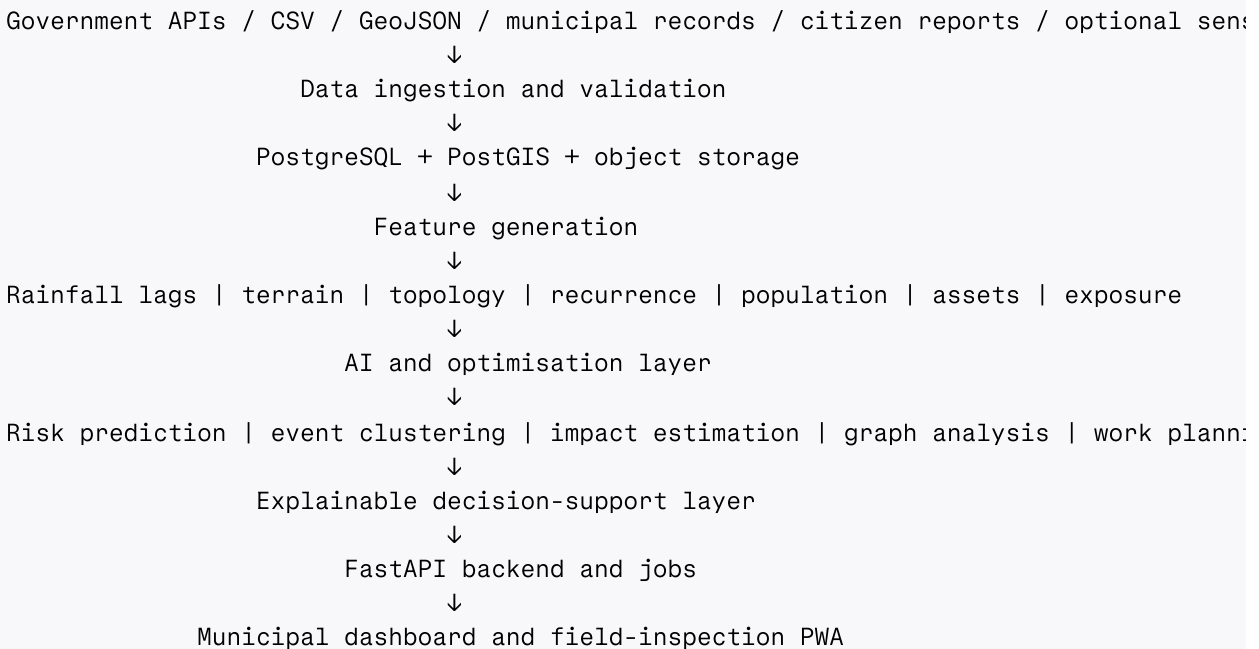
The system can identify relationships such as:

Blocked drain → waterlogging → road damage → traffic disruption → delayed emergency response.

12. Closure verification

After a reported repair, the system can compare evidence, later complaints, and recurrence to identify whether the issue was actually resolved.

Technical Architecture



Recommended Technology Stack

Frontend

- React or Next.js.
- MapLibre GL or Leaflet.
- Tailwind CSS or a professional component library.
- Time slider for forecast horizons.
- Risk, exposure, recurrence, graph, and intervention layers.

Backend

- Python.
- FastAPI.
- PostgreSQL.
- PostGIS.
- Redis and Celery, or scheduled background jobs.
- REST APIs.
- GeoJSON support.

Machine learning

- LightGBM or XGBoost for risk prediction.
- Logistic regression and seasonal-naive forecasting as baselines.
- SHAP for explanations.
- Spatial-temporal clustering for duplicate and recurrence detection.
- NetworkX or PostGIS graph operations for dependency analysis.
- OR-Tools for crew and budget optimisation.
- YOLO only for a focused image-evidence module.
- Indic ASR and multilingual NLP only if multilingual input is included.
- LLM/RAG only for structured extraction, SOP retrieval, and explanation; not for generating risk probabilities.

Deployment

- Docker.
- Docker Compose for the prototype.
- Managed PostgreSQL/PostGIS.
- Cloud deployment through a student-accessible platform.
- Scheduled data-ingestion jobs.
- Model-version storage.
- Audit logs for predictions and recommendations.

AI/ML Components

Civic issue classification

Input: Citizen text, speech transcript, image, location.

Output: Issue category, severity indicators, and confidence.

Possible method: Lightweight image model, rule-assisted NLP, and multilingual text encoder.

Metrics: Macro-F1, precision, recall, per-class recall, and confusion matrix.

Duplicate and incident clustering

Input: Location, timestamp, issue type, text, and image features.

Output: Unified civic incident clusters.

Possible method: Spatial-temporal clustering combined with text or image embeddings.

Metrics: Pairwise precision/recall and cluster purity.

Risk forecasting

Input:

- Historical incident counts.
- Rainfall totals and anomalies.
- Rainfall lags.
- Elevation and flow indicators.
- Drainage and road connectivity.
- Population exposure.
- Recurrence.
- Seasonal features.
- Infrastructure proxies.

Output: Probability or risk class for 24, 48, and 72 hours.

Recommended model: LightGBM or XGBoost.

Metrics:

- PR-AUC.
- Recall@top-k risk areas.
- Brier score.
- Calibration error.
- Time-based validation.
- Spatial holdout validation.

Asset failure prediction

Input: Asset age, type, material, history, repairs, rainfall, environment, and spatial variables.

Output: Failure hazard or vulnerability score.

Recommended model: Survival analysis or XGBoost when asset labels exist.

Metrics: Concordance index and time-dependent AUC.

Spatial risk analysis

Input: Geotagged incidents, road network, drainage network, elevation, and waterbodies.

Output: Hotspots, propagation paths, and risk corridors.

Recommended methods: KDE, Moran's I, network analysis, flow accumulation, and graph propagation.

Population and facility impact estimation

Input: Risk polygons or cells, population grids, roads, schools, hospitals, and public assets.

Output: Estimated exposed population and assets.

Method: Population-weighted spatial overlay.

Important limitation: The result is an estimated exposure value, not a verified disaster-impact count.

Work-order optimisation

Input: Predicted risk, population impact, intervention type, crew availability, skills, distance, time, cost, and ownership.

Output: Feasible inspection and maintenance schedule.

Recommended method: OR-Tools constraint programming or mixed-integer optimisation.

Metrics:

- Risk reduction per budget unit.
- Population exposure reduced.
- Travel distance saved.
- Number of critical locations addressed.
- Schedule feasibility.

India-Specific Design Principles

CivicPulse AI is designed for Indian urban conditions rather than assuming a European or American smart-city environment.

Important design principles:

- Support urban local bodies with limited staff.
- Work with incomplete and fragmented data.
- Treat sensors as optional rather than mandatory.
- Support monsoon preparedness.
- Include dense settlements and underreported locations.
- Use population exposure and critical facilities.
- Support multilingual and low-literacy reporting when needed.
- Support CSV and GeoJSON imports from legacy systems.
- Provide low-bandwidth maps and field workflows.
- Respect departmental jurisdiction.
- Provide explainability and human approval.
- Display uncertainty instead of unsupported certainty.
- Make the system useful with public data and progressively better with official data.

Proposed Data Sources

Rainfall and weather

- Open Government Data rainfall catalogue.
- IMD public API resources.
- Central Water Commission rainfall resources.
- District and station-level rainfall data.
- Historical rainfall normals and anomalies.

Geospatial data

- OpenStreetMap roads, drains, waterways, buildings, schools, hospitals, and public facilities.
- Digital elevation and terrain layers.
- Sentinel-2 multispectral imagery.
- Sentinel-1 radar imagery for flood and water extent.
- NRSC/NDEM flood hazard and inundation layers.

Population and exposure

- Census 2011 population and density data.
- Gridded population datasets.
- Schools, hospitals, transport corridors, and public facilities.

Civic incident data

- Municipal complaint records where accessible.
- BBMP Fix My Street public data.
- BBMP grievance data.
- OpenCity datasets.
- Clearly labelled synthetic or semi-synthetic incidents for prototype training and simulation.

MVP Scope

The hackathon MVP should contain:

1. One selected Indian ward or small study area.
2. One focused hazard: drainage-related waterlogging.
3. Three forecast horizons: 24, 48, and 72 hours.
4. Public rainfall, OSM, population, and civic-event data.
5. A transparent synthetic-event generator if real labels are insufficient.
6. A baseline model and an ML model.
7. Grid-level risk prediction.
8. Risk explanations using the main contributing factors.
9. Population and critical-facility exposure estimation.
10. A drainage dependency graph.
11. Top-five preventive inspection recommendations.
12. A crew-constrained work plan.
13. A rainfall and intervention scenario simulator.
14. A clear public-data, observed-data, or simulation label.
15. A human-verification warning on every recommendation.

Advanced Version

Future versions can add:

- Water-pipe failure prediction.
- Pressure and flow sensors.

- Hydraulic simulation.
- Satellite-based flood validation.
- GNN-based infrastructure dependency learning.
- Multilingual voice reporting.
- Offline field-worker application.
- Automatic before-and-after repair verification.
- Cross-city transfer learning.
- Federated learning between ULBs.
- Live work-order integration.
- Digital-twin calibration.
- Causal intervention-effect estimation.

Two-Minute Demonstration

1. Select the demonstration ward.
2. Show the existing historical complaint layer.
3. Explain that the complaint layer is reactive and incomplete.
4. Activate a 48-hour heavy-rainfall scenario.
5. Show predicted waterlogging and drainage-failure risk before new complaints appear.
6. Select the highest-risk location.
7. Display rainfall anomaly, recurrence, low elevation, drainage connectivity, population exposure, and nearby critical facilities.
8. Open the dependency graph and identify the likely upstream drain or outlet.
9. Reduce available crews from five to two.
10. Run the resource optimiser.
11. Display the selected preventive inspection order.
12. Simulate clearing one drain.
13. Show the reduction in predicted exposure and residual risk.
14. Display the final message: “AI recommendation; field verification required.”

Measurable Evaluation Metrics

Prediction

- PR-AUC.
- Recall@top-10 risk cells.
- Precision at the critical-risk threshold.
- Brier score.

- Calibration error.
- Lead time before the event.
- Performance across spatial and temporal holdouts.

Operational value

- Population exposure reduction in scenarios.
- Risk reduction per budget unit.
- Crew travel distance saved.
- Number of high-risk locations covered.
- Time required to generate a preparedness plan.
- Number of duplicate incidents merged.
- Percentage of recommendations with evidence.
- Percentage of recommendations requiring human inspection.

Fairness and reliability

- Risk coverage in high- and low-reporting areas.
- Performance across wards.
- Missing-data sensitivity.
- Forecast sensitivity to rainfall uncertainty.
- Model performance when complaint data is reduced.

Risks and Mitigation

Lack of real failure labels

Use public incidents, manual labels, and transparent semi-synthetic simulation. Evaluate on time-separated held-out events wherever possible.

Data leakage

Use time-based splits and spatial holdouts. Never randomly split repeated incidents from the same location across training and testing.

Incomplete drainage data

Treat OSM and inferred networks as proxies. Show coverage confidence and allow officers to edit or verify infrastructure relationships.

False precision

Use calibrated models, confidence bands, evidence cards, and an inspect-before-action category.

Weather forecast error

Recompute risk as forecasts change and show the weather-data confidence.

Old population data

Describe the result as estimated exposure, not current census certainty.

Complaint bias

Normalize by population, reporting coverage, and independent geographic evidence.

Overly broad scope

Build one hazard, one ward, and three decisions: where risk is highest, who is exposed, and what action should occur first.

Government adoption barriers

Support CSV/GeoJSON import, human approval, explainability, offline field inspection, and department ownership.

Privacy

Avoid exposing personal information. Aggregate reports and display only the minimum location precision needed for operations.

Hallucinated recommendations

Use deterministic action templates and verified municipal SOPs. Do not allow an LLM to invent operational instructions or risk values.

Synthetic data distrust

Label every synthetic layer and provide separate public-data, observed-data, and simulation modes.

Final Project Statement

CivicPulse AI is an explainable predictive civic infrastructure intelligence platform for Indian urban local bodies. Its focused implementation, CivicPulse Monsoon, predicts drainage and waterlogging risks 24–72 hours before likely failure by combining rainfall, historical civic incidents, terrain, geospatial infrastructure relationships, population exposure, and recurrence patterns.

The system identifies probable upstream causes, estimates the affected population and critical facilities, prioritizes preventive inspections, optimizes limited municipal crews and budgets, and allows authorities to test rainfall and intervention scenarios.

CivicPulse AI is not another complaint-management application. It is a preventive municipal decision-support system that helps authorities answer:

- Where is the next civic infrastructure problem likely to occur?
- When is it likely to become critical?
- Why is the location at risk?
- Who and what may be affected?
- Which infrastructure is probably responsible?
- Which department should act?
- What should be done first?
- What can be achieved with the available crews and budget?
- How much risk remains after the intervention?

Final Pitch

CivicPulse Monsoon predicts where drainage-linked civic failures are likely to occur before complaints, explains why, estimates who will be affected, and generates the best preventive inspection plan under municipal constraints.

Final Recommendation

MODIFY the original broad concept.

Build CivicPulse Monsoon as a focused, explainable, low-data, India-specific predictive system for drainage and waterlogging risk, infrastructure dependency analysis, impact estimation, and preventive municipal work planning.

This focused version is more technically credible, more measurable, more feasible for a student hackathon, and more differentiated from ordinary civic complaint applications.